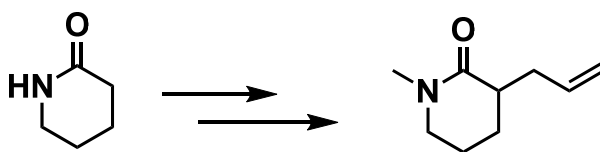
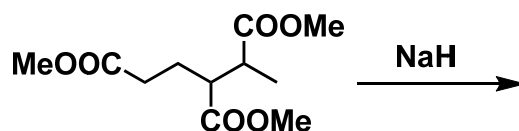


1. Complete the following transformation use necessary organic and inorganic reagents.



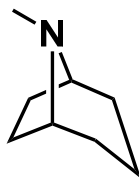
2. Draw the major product for the following reaction and suggest a mechanism.



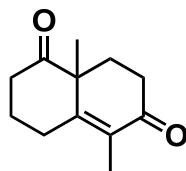
3. Draw the structures of A for the following transformation and suggest a mechanism. The NMR spectrum of A shows that two species are present. Both show two 3H triplets at about $\delta_H = 1$ and two 2H quartets at about $\delta_H = 3$ ppm. One has a very low field proton and an ABX system at 2.1–2.9 with J_{AB} 16 Hz, J_{AX} 8 Hz, and J_{BX} 4 Hz. The other has a 2H singlet at 2.28 and two protons at 5.44 and 8.86 coupled with J 13 Hz. One of these protons exchanges with D_2O . Any attempt to separate the mixture (for example by distillation or chromatography) gives the same mixture. Explain the NMR data.



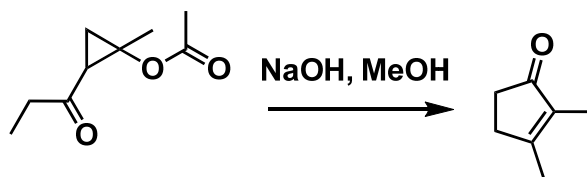
4. Using organic reagents that contain four carbon atoms or less and necessary inorganic reagents, synthesize the following compound:



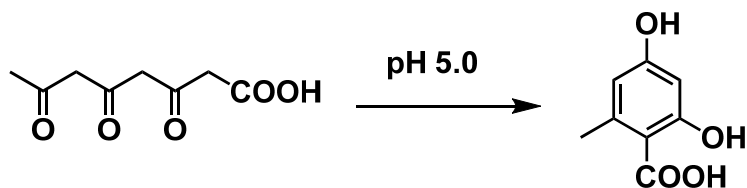
5. Using organic reagents that contain four carbon atoms or less and necessary inorganic reagents, synthesize the following compound:



6. Suggest a mechanism for the following reaction.



7. Tetraacetic acid (or a biological equivalent) is suggested as an intermediate in the biosynthesis of phenolic natural products. In the laboratory, it can be readily converted to orsellinic acid. Suggest a mechanism for this reaction.



8. Draw the structure of Compound B and suggest a mechanism.

